

BAGIAN 4

Dikarenakan fungsi $I(t)$ di sekitar $t=0$, maka deretnya dinamakan deret Maclaurin, yang merupakan deret Taylor baku

• > kondisi awal (pada $t=0$)

$$N = 100.000$$

$$\beta = 0,25$$

$$\gamma = 0,1$$

$$S(0) = 99.990$$

$$I(0) = 10$$

• > Persamaan deret Taylor untuk $I(t)$ di sekitar $t=0$ hingga orde 4

$$I(t) = I(0) + I'(0)t + \frac{I''(0)}{2!}t^2 + \frac{I'''(0)}{3!}t^3 + \frac{I^{(4)}(0)}{4!}t^4$$

→ Turunan pertama

$$I'(t) = \frac{\beta}{N} SI - \gamma I$$

$$I'(0) = \frac{0,25}{100.000} (99.990 \cdot 10) - 0,1(10) = 2,49975 - 1 = 1,49975$$

$$S'(0) = -\frac{\beta}{N} S_0 I_0 = -2,49975$$

→ Turunan kedua

$$I''(t) = \frac{\beta}{N} (S'I + SI'') = \gamma I'$$

$$I''(0) = \frac{0,25}{100.000} [(-2,49975 \times 10) + (99.990 \times 1,49975)] = 0,1(1,49975)$$

$$= 0,3748375 - 0,149975 = 0,2248625$$

$$S''(0) = -0,3748375$$

→ Turunan ketiga

$$I'''(t) = \frac{\beta}{N} (S''I + 2S'I' + SI'') - \gamma I''$$

$$= \frac{0,25}{100.000} [(-0,3748375 \times 10) + (2 \times -2,49975 \times 1,49975) + (99.990 \times 0,2248625)] - (0,1 \times 0,2248625)$$

$$= \frac{0,25}{100.000} (22,472,745) - 0,02248625 =$$

$$= 0,05618186 - 0,02248625 = 0,0336956$$

$$S'''(0) = -0,05618186$$

→ Turunan keempat

$$\begin{aligned}
 I^{(4)}(t) &= \frac{\beta}{N} (5''' I + 35'' I' + 35' I'' + 5 I''') - \gamma I^{(4)} \\
 &= \frac{0,25}{100.000} \left[(-0,05618186 \times 10) + 3(-0,7748375 \times 1,49975) \right. \\
 &\quad \left. + (3 \times -2,99975 \times 0,2248625) + (99.990 \times 0,0336956) \right] - 0,1(0,0336956) \\
 &= \frac{0,25}{100.000} (3.365,28856) - 0,00336956 = 0,00841322 - 0,00336956 \\
 &= \underline{0,00504366}
 \end{aligned}$$

$$I(t) = I(0) + I'(0)t + \frac{I''(0)}{2!} t^2 + \frac{I'''(0)}{3!} t^3 + \frac{I^{(4)}(0)}{4!} t^4$$

$$I(t) = 10 + 1,49975t + \frac{0,2248625}{2} t^2 + \frac{0,0336956}{6} t^3 + \frac{0,00504366}{24} t^4$$

$$I(t) = 10 + 1,49975t + 0,112431t^2 + 0,005616t^3 + 0,000210t^4$$